



Doxycycline to prevent bacterial sexually transmitted infections in the USA: final results from the DoxyPEP multicentre, open-label, randomised controlled trial and open-label extension

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Summary

Background Doxycycline post-exposure prophylaxis (doxy-PEP) is a promising intervention to reduce bacterial sexually transmitted infections (STIs). We evaluated the effect of doxy-PEP on STI incidence and antimicrobial resistance in men who have sex with men and transgender women for up to 12 months of follow-up, including an open-label extension.

Methods DoxyPEP, an open-label trial in Seattle (WA, USA) and San Francisco (CA, USA) among men who have sex with men and transgender women with at least one bacterial STI in the past year, randomly assigned participants by clinic (with computer-generated variable block sizes) 2:1 to doxy-PEP (200 mg doxycycline delayed-release tablets 24–72 h after condomless sex) or standard care. The independent endpoint adjudication committee was masked to group assignment. The primary outcome was presence of one or more bacterial STIs (*Neisseria gonorrhoeae*, *Chlamydia trachomatis*, or early syphilis) each quarter. This outcome was assessed in the modified intention-to-treat cohort, which included participants with at least one follow-up quarter (ie, ~3 months) in their as-randomised assignment. After early termination of the randomised phase for efficacy, all participants still enrolled were offered doxy-PEP in an open-label extension (OLE). We report quarterly incidence of bacterial STIs for the as-randomised and OLE periods. Safety was assessed in all participants with any follow-up data. The trial was registered with ClinicalTrials.gov (NCT03980223) and is completed.

Findings From Aug 19, 2020, to May 13, 2022, we enrolled 637 participants; 592 participants completed at least one follow-up quarter in the randomised phase (411 in the doxy-PEP group and 181 in the standard-care group) and 282 in the OLE phase (207 in the doxy-PEP group and 82 in the standard-care group). STIs were present in 129 (12·0%) of 1077 quarters in the doxy-PEP group versus 139 (30·5%) of 455 quarters in the standard-care group during the as-randomised period, showing an absolute difference of 19 percentage points and a relative risk of 0·39 (95% CI 0·31–0·49, $p<0·0001$). During the OLE, STIs were diagnosed in 51 (13%) of 388 quarters among those continuing doxy-PEP and 25 (17%) of 145 quarters among standard-care participants who initiated doxy-PEP. Throughout all quarters for participants on doxy-PEP, there was one grade 2 laboratory abnormality and five grade 3 adverse events that were possibly or probably related to doxy-PEP. No serious adverse events were attributed by site investigators to doxycycline. Of participants with positive gonorrhoea cultures during the study, eight (27%) of 29 taking doxy-PEP versus five (24%) of 21 not taking doxy-PEP had tetracycline resistance (minimum inhibitory concentration ≥ 2 µg/mL).

Interpretation Doxy-PEP was effective in reducing bacterial STIs in this population of men who have sex with men and transgender women, including during an open-label extension when doxy-PEP efficacy was known. Doxy-PEP was well tolerated, highly acceptable, and with no new safety signals.

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Introduction

Sexually transmitted infections (STIs) are a major public health challenge, affecting sexual and reproductive health. During the past decade, STI rates have increased in the USA and globally, disproportionately affecting specific

populations, such as men who have sex with men and transgender women.^{1,2} In the USA, syphilis rates are the highest since the 1950s, with an alarming rise in congenital syphilis. Acceptable and effective biomedical interventions are needed to reduce STI incidence.

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Research in context

Evidence before this study

Sexually transmitted infections (STIs) have been on the rise globally and there is a clear need for effective strategies to reduce STIs in populations that are disproportionately affected. Doxycycline as post-exposure prophylaxis (doxy-PEP) after condomless sex has emerged as a promising tool to reduce bacterial STIs. We searched PubMed on Nov 19, 2024, with the terms “postexposure doxycycline” and “sexually transmitted infections” for full-length articles published from the database inception to Nov 19, 2024. This search found 82 publications; of these, four were randomised controlled trials of doxycycline used as post-exposure prophylaxis. These four trials included our interim analysis published in 2023, which showed a significant reduction in bacterial STIs (80% reduction in chlamydia and syphilis and a 50% reduction in gonorrhoea) in a population of men who have sex with men and transgender women. A French study of doxy-PEP in men who have sex with men published in 2018 also showed a significant reduction in chlamydia and syphilis, but not gonorrhoea. A subsequent French trial among men who have sex with men published in 2024 showed a significant reduction in chlamydia and syphilis (83%) as well as gonorrhoea (33%) with doxy-PEP use but did not show a reduction in gonorrhoea in participants with serogroup B outer membrane vesicle meningococcal vaccination. High-level tetracycline resistance was more frequently observed in incident gonorrhoea in those taking doxy-PEP versus those not taking doxy-PEP. A trial in young women aged 18–30 years in Kenya showed no efficacy of doxy-PEP on chlamydia and gonorrhoea incidence; however, adherence to doxycycline was low.

Added value of this study

The final results of the US DoxyPEP study show the efficacy of doxycycline as post-exposure prophylaxis in a population of men who have sex with men and transgender women, including during an open-label extension when doxy-PEP efficacy was known. During as-randomised follow-up time,

at least one STI was diagnosed in 129 (12.0%) of 1077 quarters in the doxy-PEP group versus 139 (30.5%) of 455 in the standard-care group for an absolute difference of 19 percentage points and relative risk of 0.39 (95% CI 0.31–0.49, $p < 0.0001$). Doxy-PEP reduced all three bacterial STIs. This reduction was sustained during the open-label follow-up; those continuing doxy-PEP had a sustained reduction in STIs with a quarterly STI incidence of 13% and those initiating doxy-PEP had a reduction in quarterly STI incidence to 17%, despite a doubling of sexual partners and condomless anal sex. In incident gonorrhoea, resistance to the tetracycline class was similar in the doxy-PEP group (27%) and standard-care group (24%); no cephalosporin resistance was identified. We observed a reduction in *Staphylococcus aureus* colonisation in both doxy-PEP and standard-care groups with a significant increase in doxycycline resistant *S aureus* during doxy-PEP use. Doxy-PEP was well tolerated and highly acceptable, with no new safety signals.

Implications of all the available evidence

Randomised controlled trial data indicate the efficacy of doxy-PEP in men who have sex with men and transgender women in reducing chlamydia and syphilis and, to a lesser degree, gonorrhoea. Data from initial implementation studies in the USA and from US Centers for Disease Control and Prevention surveillance show a promising potential public health effect of doxy-PEP use to reduce chlamydia and syphilis in men who have sex with men. Doxy-PEP effectiveness for gonorrhoea is probably affected by community prevalence of tetracycline resistance in *Neisseria gonorrhoeae*; more effective interventions are needed for gonorrhoea. Additional studies are warranted to evaluate equitable access to doxy-PEP in populations of men who have sex with men with the highest STI burden, to examine efficacy in other populations, including cisgender women and men who have sex with women, and to monitor the long-term effect of intermittent doxycycline use on antimicrobial resistance.

Doxycycline taken after sex as post-exposure prophylaxis (doxy-PEP) is an important new biomedical intervention that has been shown to reduce *Chlamydia trachomatis* and syphilis incidence markedly and, to a lesser degree, *Neisseria gonorrhoeae* in three open-label trials among men who have sex with men and transgender women with a history of recent (ie, past 12 months) bacterial STIs.^{3–5}

The US DoxyPEP study randomly assigned men who have sex with men and transgender women to receive open-label doxy-PEP after condomless sex or standard care without doxy-PEP. In the planned interim efficacy analysis of this study, the overall reduction in bacterial STIs per quarter of doxy-PEP use met the prespecified efficacy threshold in both people living with HIV and those taking HIV pre-exposure prophylaxis (PrEP), with a relative risk (RR) of 0.38 (95% CI 0.24–0.60) for people

living with HIV and 0.34 (0.24–0.46) for people taking PrEP. DoxyPEP reduced incident *C trachomatis*, syphilis, and *N gonorrhoeae*.³

All participants were notified about the efficacy results, and those randomly assigned to standard care were offered doxy-PEP for their remaining months of study participation in an open-label extension (OLE). This extension provided an opportunity to examine doxy-PEP efficacy in the setting of participants' awareness of trial results, which potentially could affect sexual behaviour and doxy-PEP use.

This final analysis reports all participants' follow-up in the DoxyPEP study, including the OLE, and assesses efficacy, safety, antimicrobial resistance patterns in *Staphylococcus aureus* and *N gonorrhoeae*, and changes in sexual behaviour.

This manuscript was accepted before Executive Orders issued by President Trump's Administration related to gender ideology and diversity, equity, and inclusion in the USA, which have made some of the materials referenced in this Article inaccessible. The references were available upon acceptance of the manuscript.

Methods

Study design

DoxyPEP was a multicentre, open-label, randomised trial investigating 200 mg of doxycycline hyclate within 24–72 h after condomless sex to reduce the incidence of bacterial STIs, conducted at two sexual health clinics and two HIV clinics in Seattle (WA, USA) and San Francisco (CA, USA). An independent data safety monitoring board (DSMB) reviewed the safety and efficacy data and on May 13, 2022, recommended enrolments be stopped and early discontinuation of the standard-care group as the prespecified stopping boundary for effectiveness was met. The interim primary results of the DoxyPEP study have been previously published.³

After participants were notified of the study findings, those randomly assigned to standard care were offered open-label doxy-PEP. Participants who had not exited the study continued with their remaining follow-up in the OLE, up to 12 months of total study participation. The study protocol was approved by the University of California, San Francisco (San Francisco, CA, USA) institutional review board, which served as the primary institutional review board. All participants provided written informed consent in English. The trial was registered with ClinicalTrials.gov (NCT03980223).

Participants

Eligible participants were aged 18 years or older, were assigned male at birth, were living with HIV or taking or planning to start HIV PrEP, had condomless sexual contact with another person assigned male at birth in the past 12 months, and had a history of at least one bacterial STI (ie, *N gonorrhoeae*, *C trachomatis*, or early syphilis) in the past 12 months. Exclusion criteria were allergy to the tetracycline class, medications that could affect doxycycline metabolism or are contraindicated with doxycycline per prescribing information, and anticipated use of doxycycline for non-STI prevention purposes, such as for acne treatment. Study candidates were recruited from HIV and sexual health clinics through flyers and outreach to providers.

Randomisation and masking

Participants were randomly assigned 2:1 to open-label doxy-PEP or standard care without doxy-PEP. Randomisation was according to study clinic with use of computer-generated, variable block size. An independent committee adjudicated all STI endpoints and were masked to the treatment group. Interim statistical

analyses of the primary endpoint were not shared with the study investigators until after the May, 2022 DSMB meeting when recruitment was halted due to efficacy.

Procedures

Participants randomly assigned to the doxy-PEP group were provided with doxycycline in 200 mg delayed-release tablets and counselled to take 200 mg ideally within 24 h but no later than 72 h after condomless anogenital, vaginal, or oral sex and to take no more than one dose every 24 h. Participants had quarterly visits for 12 months and sexual activity, doxycycline-associated adverse events, and adherence for the previous 3 months were ascertained through computer-assisted surveys. Gender (cisgender man, transgender woman, or other), race, and ethnicity were recorded as per self-report. Quarters were categorised as occurring during the randomised phase of the study (as randomised) or the OLE. OLE quarters are divided into doxy-PEP OLE (randomly assigned initially to doxy-PEP and continued on doxy-PEP in the OLE) and standard-care OLE (randomly assigned to standard care without doxy-PEP and started doxy-PEP during the OLE, with doxycycline available for the entire quarter). Previously published results included quarters completed before May 13, 2022 (ie, the planned interim review when the DSMB recommended ending enrolment). This update includes all follow-up quarters in the study. For each participant, the visit on or after May 13, 2022, defined the beginning of the OLE follow-up, as standard-care participants were offered doxy-PEP after this date. Adherence was assessed by self-report at the quarterly visits. Safety was evaluated based on the Division of AIDS adverse event grade 2 and higher haematological and hepatic laboratory abnormalities,⁶ all grade 3 and 4 adverse events judged to be associated with doxycycline, and all serious adverse events. Sexual behaviour measurements included self-report of the number of partners and the number of condomless insertive and receptive anal sex acts.³

Gonorrhoea and chlamydia testing was done quarterly on urine and swabs from the pharynx and rectum through nucleic acid amplification testing with Aptima (Hologic, San Diego, CA, USA) or Xpert NG/CT (Cepheid, Sunnyvale, CA, USA). Blood was obtained for syphilis serologies. STI treatment was provided through local care following US Centers for Disease Control and Prevention (CDC) STI guidelines.⁷ Before treatment, participants with a positive *Neisseria gonorrhoeae* nucleic acid amplification test were asked to return for swabs for gonococcal culture with phenotypic resistance testing through CDC Strengthening the US Response to Resistant Gonorrhea and Antimicrobial Resistance Laboratory Network surveillance programmes.⁸

For evaluation of *Staphylococcus aureus*, swabs from the anterior nares and oropharynx were obtained at baseline,

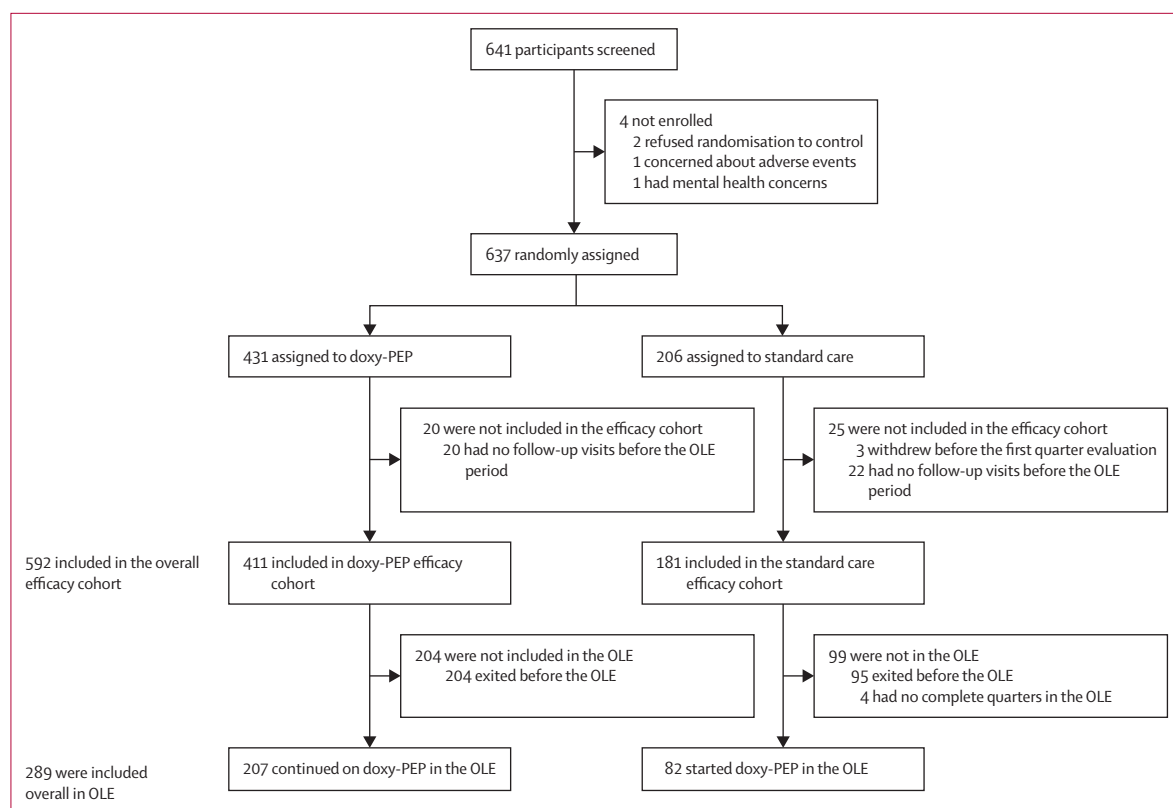


Figure 1: Trial profile

Doxy-PEP=doxycycline post-exposure prophylaxis. OLE=open-label extension.

6 months, and 12 months for culture in both groups. In an exploratory analysis, a subset of swabs from participants taking doxy-PEP with incident rectal or pharyngeal chlamydia were cultured at the University of Washington chlamydia laboratory to evaluate for phenotypic *Chlamydia trachomatis* resistance with testing in McCoy cells with a minimum inhibitory concentration (MIC) of 0.064 µg/mL.⁹

Outcomes

The primary endpoint was presence of one or more bacterial STIs (*N gonorrhoeae*, *C trachomatis*, or early syphilis) each quarter.

Secondary outcomes were incidence of each individual STI (ie, gonorrhoea, chlamydia, and syphilis); safety, tolerability, and acceptability; changes in sexual behaviour; association of doxycycline use with resistance in *N gonorrhoeae* and *C trachomatis*; and the effect of doxycycline on the gut microbiome and resistome. Assessment of *Treponema pallidum* resistance through cultures or molecular sequencing was not able to be conducted in incident syphilis cases, given the difficulty obtaining appropriate specimens and technical challenges of these laboratory techniques. History of meningococcal vaccination was assessed at baseline and at study exit, as serogroup B outer membrane vesicle

meningococcal vaccines can reduce gonorrhoea infection.^{10,11}

Safety and antimicrobial resistance (AMR) reporting included all participants during all available follow-up time. Tetracycline resistance in *N gonorrhoeae* isolated during doxy-PEP use (both in the as-randomised and OLE phases) was compared with resistance during no doxy-PEP use (people diagnosed within 2 weeks of study entry and incident cases during standard-care quarters).¹² Tetracycline resistance was defined as an MIC of at least 2 µg/mL and high-level resistance as at least 16 µg/mL, measured by the agar dilution method.¹² We conducted a randomised comparison by treatment group of *S aureus* colonisation and doxycycline resistance. We also assessed *S aureus* status within groups at enrolment compared with 6 months and 12 months; standard-care participants were censored when crossed over to doxy-PEP. Doxycycline-resistant *S aureus* was defined as an MIC of at least 16 µg/mL by ETEST (bioMérieux, Salt Lake City, UT, USA).¹³ *S aureus* swabs taken from standard-care participants after starting doxy-PEP during the OLE were not included.

Statistical analysis

The sample size calculations were quantified to power the original study groups and have been published in the

	Overall efficacy cohort		Open-label extension cohort	
	Doxy-PEP, n=411	Standard care, n=181	Doxy-PEP, n=207	Standard care to doxy-PEP, n=82
Age, years	37 (31–44)	36 (30–44)	36 (32–42)	35·5 (32–44)
Race				
White	262/397 (66%)	111/176 (63%)	123/199 (62%)	55/78 (71%)
Black	25/397 (6%)	13/176 (7%)	10/199 (5%)	6/78 (8%)
Asian or Pacific Islander	52/397 (13%)	15/176 (8%)	29/199 (15%)	3/78 (4%)
Multiple races or other	58/397 (15%)	37/176 (21%)	37/199 (19%)	14/78 (18%)
Hispanic or Latino ethnic group	106 (26%)	65 (36%)	52 (25%)	30 (37%)
Gender identity				
Man	390 (95%)	180 (99%)	194 (94%)	82 (100%)
Transgender woman or gender diverse	21 (5%)	1 (1%)	13 (6%)	0
Gender identities of sexual partners				
Only cisgender men	346/389 (89%)	149/166 (90%)	171/198 (86%)	66/76 (87%)
Both cisgender men and other identities	42/389 (11%)	17/166 (10%)	26/198 (13%)	10/76 (13%)
Annual income, US\$				
<20 000	78/410 (19%)	31/180 (17%)	30/206 (15%)	7 (9%)
20 001–50 000	123/410 (30%)	67/180 (37%)	47/206 (23%)	24 (29%)
>50 000	209/410 (51%)	82/180 (46%)	129/206 (63%)	51 (62%)
Living with HIV	124 (30%)	56 (31%)	45 (22%)	20 (24%)
STIs in the past 12 months				
Gonorrhoea	268 (65%)	130 (72%)	138 (67%)	56 (68%)
Chlamydia	248 (60%)	104 (57%)	133 (64%)	46 (56%)
Syphilis	77 (19%)	35 (19%)	35 (17%)	16 (20%)
≥2 STIs	221 (54%)	98 (54%)	111 (54%)	44 (54%)
Transactional sex during lifetime	101/406 (25%)	53/174 (30%)	48/204 (24%)	22/79 (28%)
Number of sex partners*	8 (4–18)	10 (5–18)	11 (5–22)	12 (5–20·5)
Number of insertive sex acts*	5 (3–16)	6 (3–15)	7 (3–18)	5 (3–12)
Total condomless insertive sex acts*	4028/4363 (92·3%)	1393/1500 (92·9%)	2268/2432 (93·3%)	704/729 (96·6%)
Number of receptive sex acts*	8 (4–20)	10 (5–20)	10 (4–20)	10 (6–23)
Total condomless receptive sex acts*	4805/5139 (93·5%)	1923/2118 (90·8%)	2862/2962 (96·6%)	1109/1170 (94·8%)
Number of group sex acts	1 (0–4)	1·5 (0–4)	2 (0–5)	2 (0–4)
No reported condom use*	161/258 (62%)	67/116 (58%)	82/143 (57%)	35/62 (56%)
Substance use*	236/404 (58%)	112/179 (63%)	126/203 (62%)	51 (62%)
Stimulants: methamphetamine, cocaine, or crack	123/236 (52%)	49/112 (44%)	70/126 (56%)	24 (47%)
Heroin or other opioids	12/236 (5%)	4/112 (4%)	6/126 (5%)	3 (6%)
Ecstasy, γ hydroxybutyric acid (also known as GHB), or ketamine	132/236 (56%)	61/112 (54%)	79/126 (63%)	31 (61%)
Amyl nitrates (also known as poppers)	185/236 (78%)	81/112 (72%)	100/126 (79%)	39 (76%)
Marijuana	193/236 (82%)	88/112 (79%)	101/126 (80%)	41 (80%)

Data are median (IQR), n/N (%), or n (%). Doxy-PEP=doxycycline post-exposure prophylaxis. STI=sexually transmitted infection. *In the 3 months before enrolment.

Table 1: Baseline participant characteristics

interim results.³ The efficacy cohort (ie, the modified intention-to-treat population) was all participants who completed at least one quarter of follow-up on their randomised study group (doxy-PEP or standard care). Efficacy was evaluated during as-randomised follow-up as the relative risk of any STI at quarterly visits between groups. STI incidence during as-randomised quarters was compared between groups with a modified Poisson model fitted with generalised estimating equations methods to account for repeated observations, assuming

an independent covariance structure, with city and HIV serostatus as covariates. The test for significance was a two-sided α of 0·05 with 95% CIs computed with robust standard errors. The same analysis was done for individual STIs.

The OLE was the subset of participants who were still in follow-up at the time of the May, 2022 DSMB decision and completed at least one full quarter of their randomised intervention. Standard-care participants who initiated doxy-PEP in the middle of a quarter had

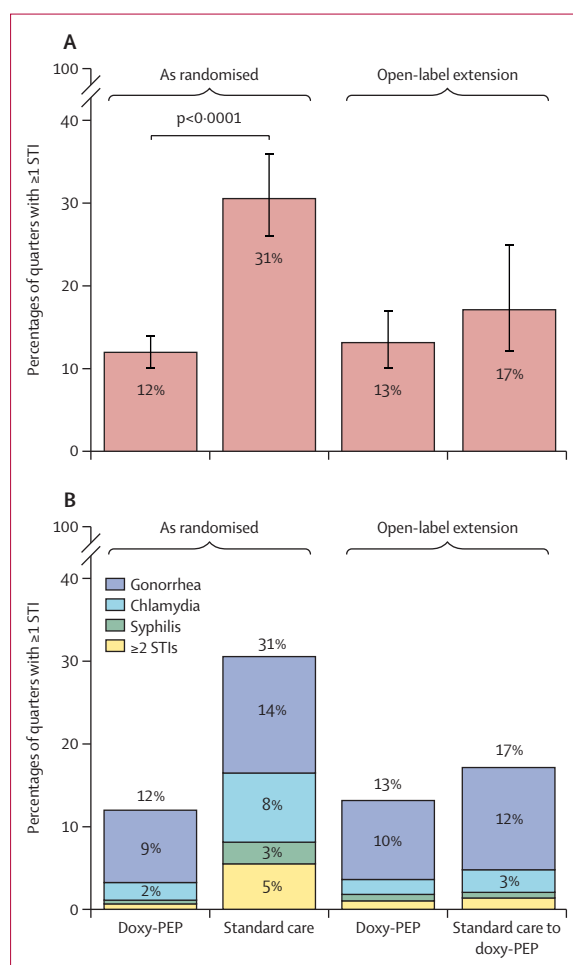


Figure 2: Incidence of STIs per quarter during as-randomised and open-label extension periods

(A) Occurrence of any incident bacterial STIs. (B) Occurrence of specific STIs. Doxy-PEP=doxycycline post-exposure prophylaxis. STI=sexually transmitted infection.

this mixed quarter excluded from analysis. Incident STIs, doxy-PEP use, and sexual behaviour in OLE quarters were reported by the study group to which participants were initially randomly assigned. No formal hypotheses were evaluated in the OLE follow-up and results were observational; our reporting is descriptive in nature.

We conducted a randomised assessment of loss of *S aureus* colonisation in participants with *S aureus* at baseline and acquisition of doxycycline-resistant *S aureus* among those without resistance at baseline. Comparison by group was restricted to the as-randomised follow-up and used Cox proportional hazards, with events assumed at time *S aureus* was evaluated. *S aureus* colonisation and doxycycline-resistant *S aureus* were also assessed cross-sectionally at each quarter of follow-up during as-randomised follow-up; we compared follow-up quarters to status at baseline within each group.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

From August 19, 2020, to May 13, 2022, the study enrolled 637 participants with 592 in the primary efficacy cohort; 45 participants did not have a full quarter in the as-randomised phase or were lost to follow-up (figure 1). Baseline characteristics are summarised in table 1. In the year before enrolment, 319 (54%) of 592 participants had two or more STIs diagnosed. In the 3 months before enrolment, participants reported a median of nine sexual partners (IQR 4–18), a median of six insertive sex acts (3–15), and nine receptive sex acts (4–20). Most reported anal sex was condomless: 5421 (92.4%) of 5863 insertive anal sex and 6728 (92.7%) of 7257 receptive anal sex acts. At enrolment, 163 of (28%) 592 had an STI diagnosed. The median time in the study, including the OLE, was 12 months (IQR 11–12.7).

299 (51%) of the 592 in the original efficacy cohort had exited the study before the May, 2022 DSMB decision. 289 (97%) of the remaining 293 participants contributed a least one full quarter of doxy-PEP use to the OLE analysis: 207 participants randomly assigned to doxy-PEP and 82 standard-care participants who chose to start doxy-PEP during the OLE (figure 1). Four standard-care participants who were still enrolled and thus eligible for the OLE did not contribute to the OLE analysis; three had only one mixed quarter (ie, started doxy-PEP halfway through the quarter) of follow-up and one declined to start doxy-PEP (all quarters for this participant were analysed as randomised). Median OLE quarters per participant was 2 (IQR 1–3). The baseline characteristics of the OLE subset were similar to the overall efficacy cohort (table 1). Participants in the doxy-PEP group reported a median of eight sex partners and in the standard-care group reported ten sex partners in the 3 months before enrolment. Those contributing to the OLE analysis reported a slightly higher median number of sex partners: 11 in the initial doxy-PEP group and 12 in the initial standard-care group (table 1).

During as-randomised follow-up time, one or more STI was diagnosed in 129 (12.0%) of 1077 quarters in the doxy-PEP group versus 139 (30.5%) of 455 quarters in the standard-care group, for an absolute difference of 19 percentage points and RR of 0.39 (95% CI 0.31–0.49, $p<0.0001$; figure 2). Of the individual STIs, gonorrhoea was diagnosed in 101 (9.4%) of 1077 quarters in the doxy-PEP group versus 88 (19.3%) of 455 quarters in the standard-care group (RR 0.46, 95% CI 0.35–0.60, $p<0.0001$; appendix p 2). There was a reduction in gonorrhoea at all anatomical sites, including pharyngeal gonorrhoea, the most common site of gonorrhoea infection in the study. Chlamydia was diagnosed in 28 (2.6%) of 1077 quarters in the doxy-PEP group versus

See Online for appendix

	As-randomised cohort		Open-label extension cohort	
	Doxy-PEP, number of quarters=1077	Standard care, number of quarters=455	Doxy-PEP, number of quarters=388	Standard care to doxy-PEP, number of quarters=146
Sex partners per quarter	10 (4–25)	8 (4–15)	12 (6–25)	16 (5–31)
Condomless insertive sex acts per quarter	5 (1–20)	4 (2–12)	8 (2–20)	8 (3–25)
Condomless receptive sex acts per quarter	8 (2–20)	5 (1–15)	10 (2–23)	10 (2–25)
Doxy doses per quarter	15 (4–30)	..	17 (7–32)	17 (5–30)
Condomless sex acts covered by doxy-PEP per quarter	27 827/33 761 (82.4%)	..	10 594/13 703 (77.3%)	5432/6683 (81.3%)
Group sex acts	2 (0–5)	1 (0–3)	2 (0–5)	2 (0–5)

Data are median (IQR) or n/N (%). Quarters were 3-month periods of follow-up. Doxy-PEP=doxycycline post-exposure prophylaxis.

Table 2: Sexual behaviour and doxy-PEP use by study group during follow-up

57 (12.5%) of 455 in the standard-care group (RR 0.20, 95% CI 0.12–0.32, $p < 0.0001$; appendix p 2). Early syphilis was diagnosed in seven (0.6%) of 1077 quarters in the doxy-PEP group versus 15 (3.3%) of 455 in the standard-care group (RR 0.20, 95% CI 0.08–0.48, $p = 0.0003$; appendix p 2).

During the OLE follow-up, during which all participants were using doxy-PEP, at least one STI was diagnosed in 51 (13%) of 388 OLE quarters in the original doxy-PEP group and 25 (17%) of 145 in the original standard-care group (figure 2). During the OLE, gonorrhoea occurred in 41 (11%) of 388 quarters in the doxy-PEP group and 20 (14%) 146 in the standard-care group, chlamydia in eight (2%) of 388 quarters in the doxy-PEP group and five (3%) of 146 in the standard-care group, and syphilis in four (1%) of 388 visits in the doxy-PEP group and one (1%) of 146 in the standard-care group (appendix p 2).

During the as-randomised period, participants reported a median of 15 doses (IQR 4–30) taken per quarter, equivalent to an estimated five doses per month (table 2). 27 827 (82.4%) of 33 761 condomless sex acts were reported as covered by doxy-PEP. Overall, 869 (91%) of 960 survey responses reported taking doxy-PEP consistently (defined as always or often) within 72 h after condomless sex. Both doxy-PEP OLE participants and standard-care OLE participants reported a median of 17 doses (IQR 7–32 for the original doxy-PEP group; IQR 5–30 for the original standard-care group) per quarter, equivalent to an estimated 5.6 doses per month. Participants reported using doxy-PEP for 10 594 (77.3%) of 13 703 condomless sex acts per quarter in the doxy-PEP OLE group and 5432 (81.3%) of 6683 in the standard-care OLE group. 331 (90%) of 369 survey responses in the doxy-PEP OLE group and 124 (88%) of 141 survey responses in the standard-care group reported taking doxy-PEP consistently (ie, always or often) within 72 h after condomless sex.

During the as-randomised period, the median sexual partners per quarter reported were ten (IQR 4–25) in the doxy-PEP group and eight (4–15) in the standard-care group. Median condomless insertive acts were five (1–20) in the doxy-PEP group and four (2–12) in the standard-care group and median condomless receptive sex acts

were eight (2–20) in the doxy-PEP group and five (IQR 1–15) in the standard-care group (table 2).

During the OLE period, the median number of sexual partners per quarter was 12 (IQR 6–25) in the doxy-PEP OLE group and 16 (5–31) in the standard-care OLE group (table 2). A median of eight insertive condomless acts (2–20 for the doxy-PEP group; 3–25 for the standard-care group) and ten receptive condomless acts (2–23 for the doxy-PEP group; 2–25 for the standard-care group) were reported during the OLE quarterly visits (table 2).

During all quarters for participants on doxy-PEP (as-randomised and OLE periods), there was one grade 2 laboratory abnormality (elevated alanine aminotransferase) and five grade 3 adverse events (three diarrhoeal events and two headaches or migraines) that were possibly or probably related to doxycycline, all of which occurred during the as-randomised phase. No adverse events were attributed to doxy-PEP during the OLE quarters. No serious adverse events were attributed by site investigators to doxycycline (appendix p 4). 12 participants permanently discontinued doxy-PEP, six due to adverse events and six due to participant preference. No discontinuations occurred during the OLE period. During as-randomised follow-up, the observed difference in mean absolute annualised weight change was 0.8 kg (95% CI –1.0 to 2.7) in the doxy-PEP group versus 1.8 kg (0.3 to 3.4) in the standard-care group. During as-randomised doxy-PEP use, 159 (89%) of 178 of participants reported that doxy-PEP was acceptable or very acceptable; this proportion was similar to acceptability during the OLE period, with 157 (93%) of 169 of participants initially assigned to doxy-PEP and 68 (91%) of 75 newly initiating doxy-PEP in the OLE reporting doxy-PEP to be acceptable or very acceptable.

During as-randomised and OLE follow-up, 69 (18%) of 377 *N gonorrhoeae* diagnoses had results for phenotypic resistance testing: 19 at baseline, 21 during standard-care, and 29 during doxy-PEP use (both for as-randomised and OLE periods; figure 3). Culture availability was limited by samples not being collected before treatment in more than half of participants and low culture yield from rectal and pharyngeal extragenital

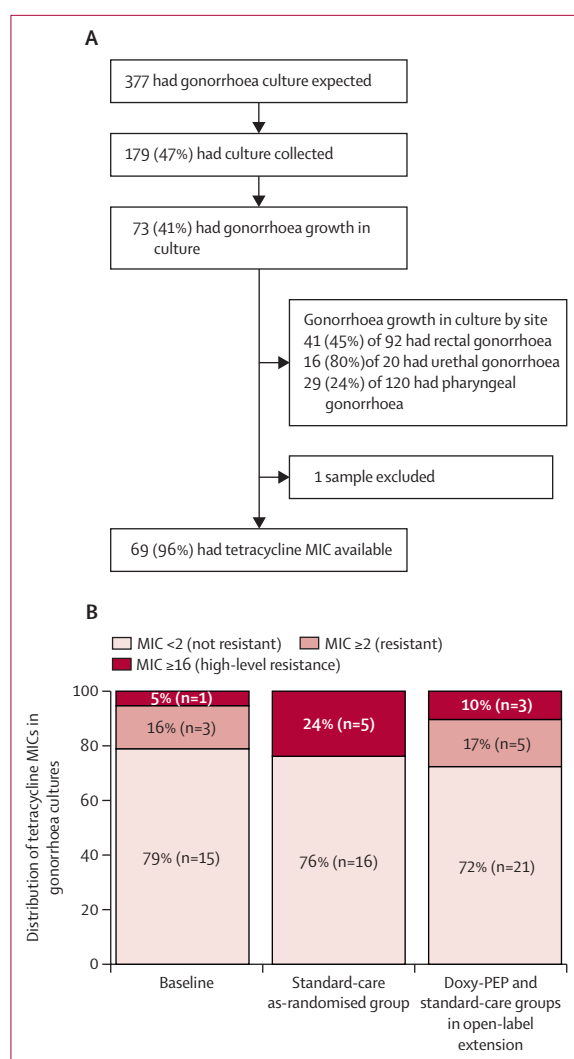


Figure 3: Available *Neisseria gonorrhoeae* culture and phenotypic antimicrobial resistance testing from gonorrhoea endpoints
Doxy-PEP=doxycycline post-exposure prophylaxis. MIC=minimum inhibitory concentration.

sites, as has been reported previously.¹⁴ At baseline, tetracycline resistance (MIC ≥2) was present in four (21%) of 19 available gonorrhoea cultures, with high-level tetracycline resistance (MIC ≥16) in one (5%) of 19 cultures. During quarters for doxy-PEP use, eight (27%) of 29 *N. gonorrhoeae* cultures had tetracycline resistance, and three (10%) of 29 had high-level resistance. During standard-care quarters, five (24%) of 21 cultures had tetracycline resistance, all high level. A quarter of isolates had at least two classes of AMR; tetracycline resistance was accompanied by other antibiotic class resistance in eight (28%) of 30 *N. gonorrhoeae* cultures during doxy-PEP use and nine (23%) of 39 cultures without doxy-PEP use (appendix pp 6–7). No cephalosporin resistance was identified.

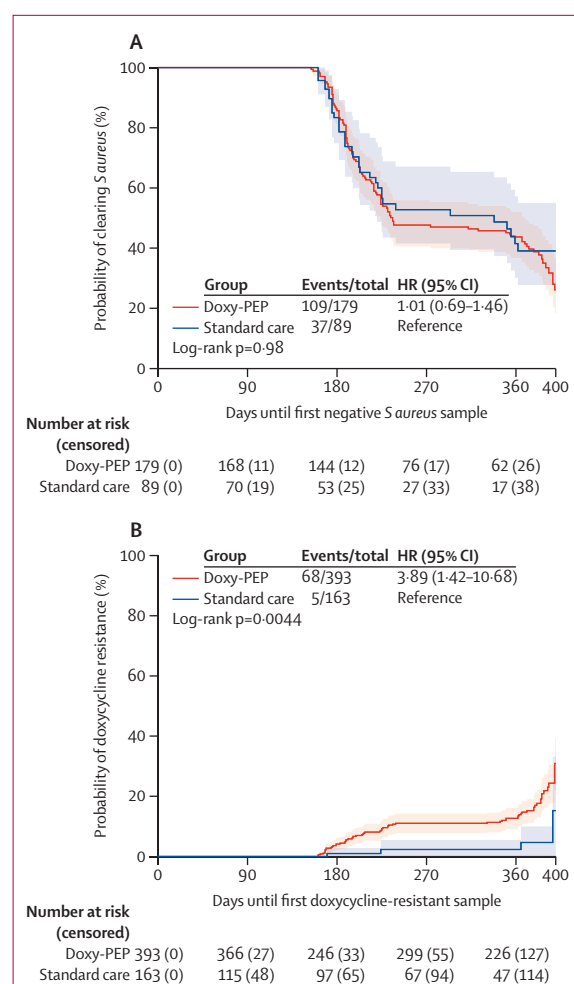


Figure 4: *Staphylococcus aureus* colonisation and development of doxycycline resistance

(A) Time to loss of colonisation in participants with *S. aureus* detected at baseline. (B) Time to detection of doxycycline-resistant *S. aureus* in those without doxycycline-resistant *S. aureus* at baseline. Cultures from standard-care participants after crossover to doxy-PEP during the open-label extension were not included. Doxy-PEP=doxycycline post-exposure prophylaxis. HR=hazard ratio.

At baseline, *S. aureus* was isolated from the oronasopharynx in 285 (45%) of all 636 sampled participants. In the subsequent analyses, standard-care participants were censored when they crossed over to doxy-PEP. In a survival analysis of those with *S. aureus* colonisation at baseline, participants taking doxy-PEP were not significantly more likely to clear *S. aureus* colonisation than those with no doxy-PEP use (hazard ratio [HR] 1.01, 95% CI 0.69–1.46, p=0.98; figure 4). Notably, a decrease in colonisation occurred in both groups over time. For those in the doxy-PEP group, 187 (43%) of 430 participants had *S. aureus* detected at baseline compared with 121 (36%) of 340 at month 12 (appendix p 10). The standard-care group also had some decline in *S. aureus* detection, from 98 (48%) of 206 participants at baseline to 42 (35%) of 120 at month 6

and 32 (42%) of 76 at month 12. Note that only 76 (37%) of 206 standard-care participants had *S aureus* culture results at 12 months primarily due to cross-over to doxy-PEP use.

Doxycycline-resistant *S aureus* was cultured from 42 (7%) of 636 participants at baseline. In a survival analysis of those without doxycycline-resistant *S aureus* at baseline, doxy-PEP use was associated with a statistically significant increase in detection of doxycycline-resistant *S aureus* (HR 3.89, 95% CI 1.42–10.68, $p=0.0044$; figure 4). During as-randomised and open-label follow-up of those who were assigned to doxy-PEP (including those with doxycycline-resistant *S aureus* at baseline), 52 (15%) of 340 doxy-PEP participants had doxycycline-resistant isolates at month 12 (appendix p 10).

At study enrolment, methicillin-resistant *S aureus* (MRSA) was present in 36 (6%) of 636 participant samples (appendix p 11). There was no change in MRSA colonisation in either group over time: in the doxy-PEP group there was MRSA colonisation in 24 (6%) of 430 participants at baseline versus 21 (6%) of 340 at month 12, and in the standard-care group there was colonisation in 12 (6%) of 206 participants at baseline and four (5%) of 76 at month 12. Doxycycline-resistant MRSA was present in eight (1%) of 636 isolates at baseline, and at month 12 there were six (2%) of 340 in the doxy-PEP group and one (1%) of 76 in the standard-care group. Of note, there were only seven total doxycycline-resistant MRSA isolates identified at month 12.

Of 16 participants taking doxy-PEP with incident *C trachomatis* who had swabs collected for chlamydia culture, five (31%) had growth in culture; all were susceptible to tetracycline.

Meningococcal vaccination status was recorded; four participants reported previous meningococcal B vaccination at baseline, and four additional participants reported meningococcal B vaccination by study month 12.

Discussion

Doxy-PEP taken within 72 h after condomless sex significantly decreased incident bacterial STIs (RR 0.39, 95% CI 0.31–0.49, $p<0.001$) among men who have sex with men and transgender women with a recent (ie, within the past year) history of STIs, based on the analysis of the efficacy cohort of participants assigned to doxy-PEP compared with the standard-care group. Doxy-PEP use was associated with a significant reduction in chlamydia (RR 0.20, 95% CI 0.12–0.32, $p<0.0001$), syphilis (0.20, 0.08–0.48, $p=0.0003$), and gonorrhoea (0.46, 0.35–0.60, $p<0.0001$). During the OLE period in the setting of known efficacy from study interim results, doxy-PEP use was associated with sustained decreases in STI incidence in those who continued doxy-PEP and a similar reduction in STIs among those who started doxy-PEP. Importantly, decreased STI rates were observed with doxy-PEP use despite reported increases in sexual partners and in

condomless sex during the OLE period for all participants and a doubling of insertive and receptive sex acts in standard-care participants initiating doxy-PEP. Given the median follow-up of two quarters during OLE, we cannot address whether these changes in sexual behaviour are temporary. However, the decreased STI rates during the OLE and slight increased number of doxycycline doses, from 15 to 17 per quarter (with a wide IQR), suggest that doxy-PEP use mitigated the STI risk despite reported increased sexual activity during the OLE follow-up. Overall, the effect of doxy-PEP on STI reduction during the OLE was consistent with results during the as-randomised period, both for each STI and by anatomical site of infection. There were no major safety or tolerability issues, and acceptability was high.

The efficacy of doxy-PEP for chlamydia and syphilis reduction has been consistently high in the three randomised controlled trials among men who have sex with men and transgender women.^{3,4,5} However, the efficacy of doxy-PEP for gonorrhoea has been lower and less consistent, which could be affected by the local prevalence of tetracycline-resistant gonorrhoea, in addition to other factors. In the current trial, doxy-PEP resulted in a 54% lower *N gonorrhoeae* incidence, within a context of *N gonorrhoeae* tetracycline resistance (MIC 2–8) of 5–18% and 10–15% high-level resistance (MIC ≥ 16) according to US CDC surveillance in 2019–22.¹⁵ In contrast, the French DOXYVAC study found a lower effectiveness of doxy-PEP against a first episode of *N gonorrhoeae*, with an adjusted hazard ratio of 0.67 ($p=0.0025$) for the time to first episode of gonorrhoea (adjusted HR 0.67, $p=0.0025$), in the national context of 75% of *N gonorrhoeae* isolates with tetracycline resistance (MIC ≥ 2).¹⁶ Although the tetracycline-resistance threshold above which doxy-PEP is ineffective to prevent gonorrhoea remains unknown, it is probable that reported increasing rates of tetracycline-resistant *N gonorrhoeae*^{17,18} will decrease doxy-PEP effectiveness against *N gonorrhoeae*. Providers should counsel patients that doxy-PEP is less effective in preventing gonorrhoea than syphilis or chlamydia. Other tools are needed to reduce the risk of gonorrhoea infection.

Intermittent, long-term use of doxycycline raises concerns about development of AMR both in bacterial STIs and colonising, off-target bacteria with pathogenic potential. With *N gonorrhoeae*, there is concern both for tetracycline and multiclass AMR. However, in this study, there were no cephalosporin-resistant *N gonorrhoeae* isolates, and multiclass resistance was similar among participants with and without doxycycline exposure. Although there is theoretical concern for tetracycline class resistance in *T pallidum* and *C trachomatis*, such resistance has not occurred despite decades of tetracycline use. However, it is important to remain vigilant for this possibility, given the widespread use of tetracyclines for many indications, including treatment of syphilis and chlamydia. The effect of doxy-PEP on *S aureus* was mixed

with no evidence of effect on loss of colonisation but a significant increase in doxycycline resistance in the doxy-PEP group. MRSA colonisation was uncommon and did not appear to be affected by doxycycline use. Notably, there was a decline in *S aureus* isolation over time in both groups. Changes in the rectal microbiome in the DoxyPEP trial have been reported in depth elsewhere; doxy-PEP use was associated with an increase in the expression of tetracycline resistance genes (but not other antimicrobial classes) but was not associated with changes in α or β diversity associated with doxy-PEP use.¹⁹

The DoxyPEP study was limited by only a year of follow-up and a median of two quarters of use during the OLE. The OLE follow-up occurred at a later timepoint than much of the as-randomised follow-up. We cannot assess temporal contributions to sexual behaviour changes seen during the OLE period, including intercurrent events, such as waves of COVID-19 and mpox emergence, which could have affected sexual behaviour. Larger cohorts of doxy-PEP users from geographically diverse settings will need to be studied for longer to understand sustained effectiveness and patterns of AMR, as well as the potential effect on sexual behaviour.

In summary, the high effectiveness, safety, tolerability, and acceptability of doxy-PEP support the US CDC recommendations to counsel men who have sex with men and transgender women with a recent history of bacterial STIs about doxy-PEP as an STI prevention tool.²⁰ Evaluation of AMR in *N gonorrhoeae* and common colonising bacteria, such as *S aureus* and *Streptococcus pneumoniae*, is warranted as part of public health surveillance and research with long-term doxy-PEP use.

Contributors

AFL and CCE wrote the manuscript. CG and DD did the statistical analysis and directly accessed and verified the underlying data. SEC, JCD, CB, CCA, CM, RP, MN, CL, RJS, EV, SPB, HS, EDC, DVH, and OOS provided input into the design, conduct, analysis, and interpretation of the study and edited manuscript drafts. GH provided input on interpretation and editing of the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

AFL reports research grants paid to her institution from Gilead, GSK, Merck, the US National Institutes of Health, and ViiV Healthcare; provision of laboratory kits and supplies from Cepheid and Hologic; and receipt of the study drug from Mayne Pharma. DD, SEC, and JCD report provision of laboratory kits and supplies from Cepheid and Hologic and receipt of the study drug from Mayne Pharma. SPB reports research grants paid to her institution from Gilead, GSK, Merck, and ViiV Healthcare. DVH and EDC report research grants from the US National Institutes of Health. CCE received honoraria from Gilead and Merck as a scientific advisor, unrelated to this research; received research grants from the US National Institutes of Health and the Bill & Melinda Gates Foundation; received provision of laboratory kits and supplies from Cepheid and Hologic; and received the study drug from Mayne Pharma. All other authors declare no competing interests.

Data sharing

The data that support the findings of this study are available from the corresponding author upon reasonable request to icrc@uw.edu, which will be reviewed by a central publications and manuscript committee. The study protocol (including informed consent) and statistical analysis plans have been made available in a previous publication.³

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References

- 1 US Centers for Disease Control and Prevention. Sexually transmitted infections surveillance, 2023. Nov 14, 2024. <https://www.cdc.gov/sti-statistics/annual/index.html> (accessed Nov 19, 2024).
- 2 WHO. Implementing the global health sector strategies on HIV, viral hepatitis and sexually transmitted infections, 2022–2030: report on progress and gaps 2024. World Health Organization, 2024.
- 3 Luetkemeyer AF, Donnell D, Dombrowski JC, et al. Postexposure doxycycline to prevent bacterial sexually transmitted infections. *N Engl J Med* 2023; **388**: 1296–306.
- 4 Molina JM, Charreau I, Chidiac C, et al. Post-exposure prophylaxis with doxycycline to prevent sexually transmitted infections in men who have sex with men: an open-label randomised substudy of the ANRS IPERGAY trial. *Lancet Infect Dis* 2018; **18**: 308–17.
- 5 Molina J-M, Bercot B, Assoumou L, et al. Doxycycline prophylaxis and meningococcal group B vaccine to prevent bacterial sexually transmitted infections in France (ANRS 174 DOXYVAC): a multicentre, open-label, randomised trial with a 2×2 factorial design. *Lancet Infect Dis* 2024; **24**: 1093–104.
- 6 US National Institute of Allergy and Infectious Diseases. DAIDS table for grading the severity of adult and pediatric adverse events, corrected version 2.1. 2017 <https://rsc.niaid.nih.gov/clinical-research-sites/daids-adverse-event-grading-tables> (accessed Nov 1, 2024).
- 7 Workowski KA, Bachmann LH, Chan PA, et al. Sexually transmitted infections treatment guidelines, 2021. *MMWR Recomm Rep* 2021; **70**: 1–187.
- 8 US Centers for Disease Control and Prevention. Combating the threat of antibiotic-resistant gonorrhea 2021. 2022. <https://www.cdc.gov/std/gonorrhea/arg/carb.htm> (accessed Oct 15, 2024).
- 9 Suchland RJ, Geisler WM, Stamm WE. Methodologies and cell lines used for antimicrobial susceptibility testing of *Chlamydia* spp. *Antimicrob Agents Chemother* 2003; **47**: 636–42.
- 10 Abara WE, Bernstein KT, Lewis FMT, et al. Effectiveness of a serogroup B outer membrane vesicle meningococcal vaccine against gonorrhoea: a retrospective observational study. *Lancet Infect Dis* 2022; **22**: 1021–29.
- 11 Petousis-Harris H, Paynter J, Morgan J, et al. Effectiveness of a group B outer membrane vesicle meningococcal vaccine against gonorrhoea in New Zealand: a retrospective case-control study. *Lancet* 2017; **390**: 1603–10.
- 12 CDC. Agar dilution antimicrobial susceptibility testing 2013. <https://www.cdc.gov/std/gonorrhea/lab/agar.htm> (accessed Oct 15, 2024).
- 13 Clinical & Laboratory Standards Institute. M100—performance standards for antimicrobial susceptibility testing, 31st edition. <https://clsi.org/about/press-releases/clsi-publishes-m100-performance-standards-for-antimicrobial-susceptibility-testing-31st-edition/#> (accessed Sept 1, 2024).
- 14 Nash EE, Pham CD, Raphael B, et al. Impact of anatomic site, specimen collection timing, and patient symptom status on *Neisseria gonorrhoeae* culture recovery. *Sex Transm Dis* 2021; **48** (12S suppl 2): S151–56.
- 15 CDC. Sexually Transmitted Disease Surveillance 2022: gonococcal isolate surveillance project national profile. US Department of Health and Human Services, 2024.
- 16 Bebear C, Bercot B, Dupin N. National Reference Centre, Bacterial Sexually Transmitted Infections Report 2017–2021. 2022. <https://www.cnr-ist.fr/ressources/editeur/CNR-IST-Bilan-2017-2021.pdf> (accessed Feb 24, 2025).

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- 17 Soge OO, Thibault C, Cannon C, Dombrowski JC, Golden MR. Trends in *Neisseria gonorrhoeae* tetracycline resistance before and after doxy-PEP implementation in Seattle, Washington. STI Prevention Conference; Sept 19, 2024 (abstr O15.3).
 - 18 Kohn R, Nguyen T, Bacon O, Sankaran M, Masinde GSC. Increasing prevalence of tetracycline-resistant gonorrhea among sexually transmitted infection clinic patients is correlated with increasing use of doxycycline for treatment and post-exposure prophylaxis (Doxy-PEP), San Francisco, 2002–2023. STI Prevention Conference; Sept 19, 2024 (abstr O15.4).
 - 19 Chu VT, Glascock A, Donnell D, et al. Impact of doxycycline post-exposure prophylaxis for sexually transmitted infections on the gut microbiome and antimicrobial resistance. *Nat Med* 2025; **31**: 207–17.
 - 20 Bachmann LH, Barbee LA, Chan P, et al. CDC clinical guidelines on the use of doxycycline postexposure prophylaxis for bacterial sexually transmitted infection prevention, United States, 2024. *MMWR Recomm Rep* 2024; **73**: 1–8.